AM5011 · 1ST REVIEW

ADAPTIVE VISUAL ODOMETRY VIA RL

Replicating and extending “Reinforcement Learning Meets Visual Odometry” (Messikommer et al., ECCV 2024)

LAVLIN JAISON · ME23B250 · OCTOBER 2026

LITERATURE REVIEW

2014–17

SVO · Forster et al.

Fast semi-direct VO. When to add a keyframe is decided by hand-tuned rules and thresholds.

2024

Messikommer et al. · ECCV

PPO agent picks keyframes and feature-grid size inside SVO; reports lower trajectory error than the rules.

2026

Nascivera et al.

Image-conditioned RL tunes frontend thresholds online; up to 8% lower ATE than static settings.

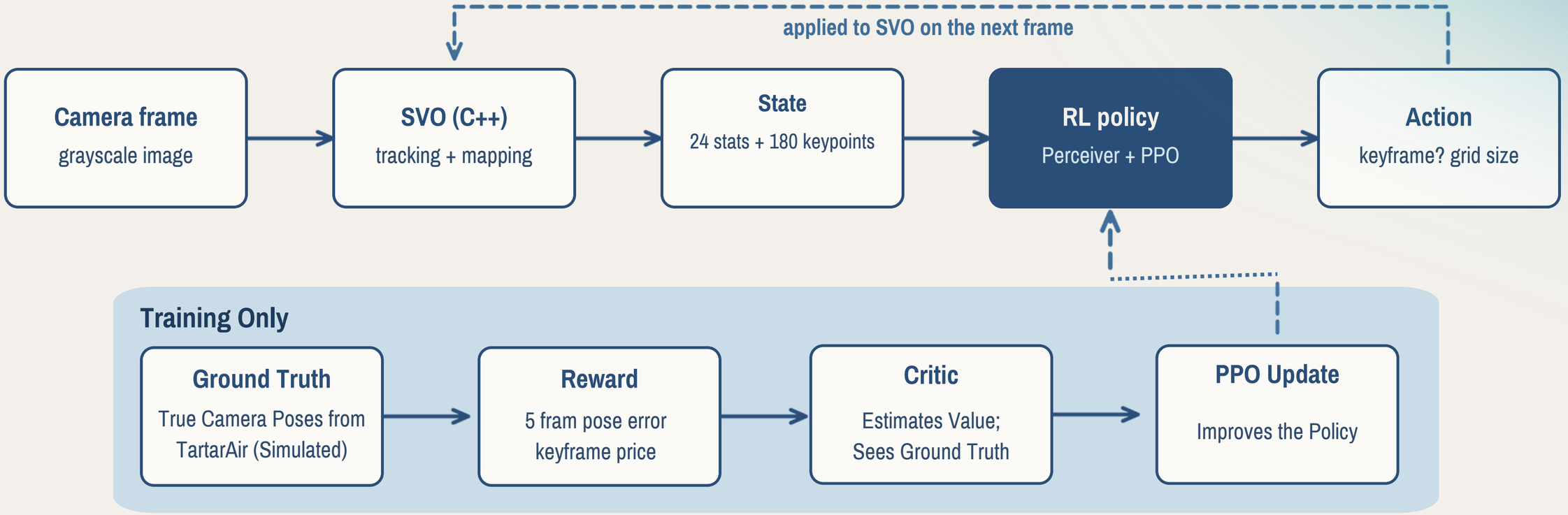
2026

Dai et al. · Pan et al.

RL keyframe selection for learned VO; cost-aware dual-agent RL for visual-inertial odometry.

Gap: no independent replication, and no test of whether RL beats a well-tuned rule set at a student's compute budget.

PROJECT ARCHITECTURE



01 · Gym environment

12 SVO instances in parallel on TartanAir; the agent acts only while SVO is tracking.

02 · Policy

Perceiver cross-attention encoder, PPO, and a critic that sees ground truth during training.

03 · Reward

Sliding-window pose error minus a keyframe cost. Ours: a shadow-SVO baseline and a keyframe budget.

RESOURCES NEEDED

No hardware components or headset: this is a software project that runs on one laptop.

Limit: the authors trained with 100 environments for 25M steps. We train with 12 for 0.4M per run, about 1.6% of their budget.

01

Compute

Apple M3 Pro laptop, CPU only. 12 parallel SVOs ≈ 550 steps/s: one 400k-step run ≈ 14 min.

02

Datasets

TartanAir (training, synthetic) · TUM-RGBD (development) · EuRoC (held-out test, download pending).

03

Software

Authors' open-source RL-VO code (GPL-3.0), SVO patched for Apple Silicon, PyTorch + Stable-Baselines3.

PROGRESS SINCE 0TH REVIEW

Phase 1

DONE

Environment setup

SVO built natively on the Mac; Gym wrapper matches the authors' env. SVO baseline within 8% of the paper.

Phase 2

DONE

Policy design

Perceiver encoder, PPO, sliding-window reward, plus 11 switchable modifications.

Phase 3

DONE

Policy training

Authors' PPO and 6 variants, 3 seeds each, with the privileged critic on TartanAir.

Phase 4

IN PROGRESS

Evaluation

TUM-RGBD and TartanAir done. EuRoC test next; KITTI if time allows.

RL METHODS TRIED

Method	Seeds	Error vs rules	TUM fails/run	TartanAir fails / traj	Keyframe rate	Verdict
SVO rules (Baseline)	-	1.00	3.0	1.5	0.29	Reference
Authors's PPO	3	1.13	2.8	1.35	0.34	13% worse
Reward tweaks (γ, window, penalties)	1-3	-	-	-	0 or 1	Keyframe Collapse
Behaviour cloning → PPO	3	1.13	4.0	1.87	0.40	Worse, less robust
Residual RL over the rules	3	1.01	3.0	1.52	0.29	Copies the rules
Constrained PPO (kf budget)	3	1.00	5.5	1.20	0.27	Tie
Shadow-relative reward	1	1.36	3.4	1.00	0.85	Cheats with keyframes
Shadow reward + kf budget	3	0.99	3.7	1.19	0.23	Best: ties the rules
+ 20-frame window (1M steps)	1	1.09	5.7	2.10	0.28	More failures

FINDINGS & NEXT STEPS

0.99

Best RL method vs SVO’s tuned rules (per-sequence error ratio, 3 seeds): a tie, not a win.

~8%

Most that any keyframe/grid choice can gain per sequence, measured on held-out SVO runs.

±7%

Noise between training checkpoints: one checkpoint can show a “win” that isn’t real.

NEXT

- EuRoC test set
- Longer training of the best method
- Write-up